

Design and Validation of Digital Holographic Microscopy Using an Off-Axis Mach–Zehnder Interferometer

Koji Abel, Jeremy Baden, and Jarek Escobar

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Abstract

Digital holographic microscopy (DHM) uses interference between a reference beam and an object beam to encode both amplitude and phase information in a single intensity image. In this capstone project, we designed, built, and iteratively refined an off-axis Mach–Zehnder interferometer using a green diode laser, two beam splitters, mirrors, a diverging lens, a converging lens, and a camera. Several initial configurations failed because the Fourier-domain first orders were either not visible, overlapped strongly with the DC term, or were dominated by specular and wavefront artifacts. The final configuration produced stable interference fringes and allowed a qualitative reconstruction of a transparent glass sample.

As part of the system characterization, a Gaussian fit to the measured laser spectrum gave a central wavelength

$$\mu = (530.006 \pm 0.004) \text{ nm}, \quad \sigma = (0.647 \pm 0.003) \text{ nm},$$

corresponding to a full width at half maximum of

$$\Delta\lambda_{\text{FWHM}} = (1.524 \pm 0.006) \text{ nm}.$$

Using the usual bandwidth-based estimate gives a coherence length of approximately

$$L_c \approx \frac{\lambda^2}{\Delta\lambda} = (0.184 \pm 0.001) \text{ mm}.$$

This value is short compared with the path-length changes present in the experiment and should therefore be treated cautiously. The spectral sampling, fit model, and spectrometer response were not characterized well enough to treat the resulting coherence length as a precision measurement. Experimentally, the strong sensitivity of the fringes to small path-length and alignment changes nevertheless showed that coherence remained one of the dominant constraints on the system.

The final reconstruction showed a clear transmission change and phase-related contrast where the diffuse glass sample intersected the object beam. The experiment therefore demonstrated the core DHM pipeline—interferometric encoding, Fourier-order isolation, complex-field reconstruction, and qualitative sample validation—with the available Physics 5CL equipment, while also identifying the main optical and computational limitations that would need to be improved for true quantitative phase microscopy.

1 Introduction and Project Scope

The reason behind our choice of a DHM capstone was to use our knowledge of optics from previous labs and courses to see how far we could push the available 5CL equipment toward a real holographic imaging system. Rather than reproducing a single prescribed setup, the project became

an iterative design problem: we needed to generate a stable hologram, determine which optical geometry produced separable Fourier orders, suppress artifacts strongly enough to reconstruct the field, and then test whether the reconstruction responded to a physical sample in a sensible way.

The goal of the project was therefore to construct a working digital holographic microscopy system and use it to image a transparent object. Unlike ordinary camera imaging, which records only intensity, DHM uses an interference pattern to retain information about the relative optical phase. Under the appropriate assumptions, the recovered phase can be related to optical path length and, for sufficiently thin and well-characterized samples, to relative height or thickness.

Our specific objectives were:

1. Build and iteratively refine an interferometer that produces visible and stable fringes,
2. Record off-axis holograms with a camera,
3. Isolate the first diffraction order in Fourier space,
4. Reconstruct relative amplitude and phase information,
5. Validate that a reconstructed feature tracks a real physical sample rather than a numerical artifact, and
6. Identify the dominant optical, mechanical, and computational limitations of the system.

The final objective became especially important because DHM is extremely sensitive to coherence, alignment, wavefront quality, camera geometry, and the details of Fourier filtering. A major part of this report is therefore not only the final reconstructed image, but the design process that led to a viable configuration: why several geometries failed, what those failures revealed about the system, why the final Mach-Zehnder configuration worked better, and how we validated the reconstruction once usable holograms were obtained.

2 Theory

2.1 Interference and Hologram Formation

Consider a monochromatic laser beam of wavelength λ . A beam splitter divides the beam into an object beam and a reference beam. We write the reference and object electric fields as complex scalar fields,

$$E_R(\mathbf{r}) = A_R(\mathbf{r})e^{i\phi_R(\mathbf{r})}, \quad E_O(\mathbf{r}) = A_O(\mathbf{r})e^{i\phi_O(\mathbf{r})}. \quad (1)$$

The camera records only the intensity of the superposed fields. Therefore, at the recording plane the measured quantity is

$$I(\mathbf{r}) = |E_R + E_O|^2. \quad (2)$$

Expanding gives

$$I = |E_R|^2 + |E_O|^2 + E_R E_O^* + E_R^* E_O. \quad (3)$$

The first two terms represent the separate intensities of the reference and object fields. They are generally concentrated near the DC region of Fourier space. The last two terms are the interference, or cross, terms. Up to conjugation, these terms contain products of the object and reference fields and therefore encode their relative phase,

$$\Delta\phi = \phi_O - \phi_R. \quad (4)$$

This is the central idea behind DHM. The camera still records only an intensity image, but the interference pattern allows information about the phase difference between the two optical paths to be recovered computationally.

2.2 Off-Axis Fourier Reconstruction

In an off-axis holography setup, the reference and object beams reach the camera with a small relative angle. That relative angle introduces a spatial carrier frequency into the interference terms. A simple plane-wave model illustrates this. If the reference field is taken to be nearly normal to the camera plane, while the object field has a small transverse wavevector, the fields can be written schematically as

$$E_R(x, y, t) = A_R e^{i(-\omega t + \phi_R)}, \quad (5)$$

and

$$E_O(x, y, t) = A_O e^{i(k_x x + k_y y - \omega t + \phi_O)}. \quad (6)$$

The transverse phase

$$k_x x + k_y y \quad (7)$$

causes the object-reference cross term to oscillate spatially across the sensor. In other words, the small angular difference between the two beams moves the interference information away from zero spatial frequency.

Taking the Fourier transform of Eq. (3) separates the signal into a central contribution and two conjugate first-order sidebands:

$$\mathcal{F}\{I\} = \mathcal{F}\{|E_R|^2 + |E_O|^2\} + \mathcal{F}\{E_R E_O^*\} + \mathcal{F}\{E_R^* E_O\}. \quad (8)$$

In the ideal plane-wave model,

$$|E_R|^2 = |A_R|^2, \quad |E_O|^2 = |A_O|^2, \quad (9)$$

so the separate beam intensities do not contain transverse spatial frequency and would contribute only at the center of Fourier space. One of the ± 1 sidebands could then be isolated, shifted to the origin, and inverse transformed to recover the object field relative to the reference field, up to conjugation and an overall phase factor.

Experimentally, our beams were not ideal plane waves. The expanded laser beam had a nonuniform profile and residual wavefront curvature, while reflections from lenses and other optical surfaces produced additional interference structure. One visible example was Newton's rings, shown in Fig. 1. These effects cause the individual object and reference intensities themselves to contain spatial structure. As a result, the central Fourier content can spread outward and partially obscure the first-order sidebands.

For this reason, we introduced an additional background-subtraction step. We recorded the object beam by itself, the reference beam by itself, and the full hologram with both beams present. If

$$I_R = |E_R|^2, \quad I_O = |E_O|^2, \quad (10)$$

then the background-subtracted hologram is approximately

$$I_f = I - I_R - I_O \approx E_R E_O^* + E_R^* E_O. \quad (11)$$

Its Fourier transform is therefore dominated by the two interference sidebands:

$$\mathcal{F}\{I_f\} \approx \mathcal{F}\{E_R E_O^*\} + \mathcal{F}\{E_R^* E_O\}. \quad (12)$$

We summarize the reconstruction process used throughout the project as follows:

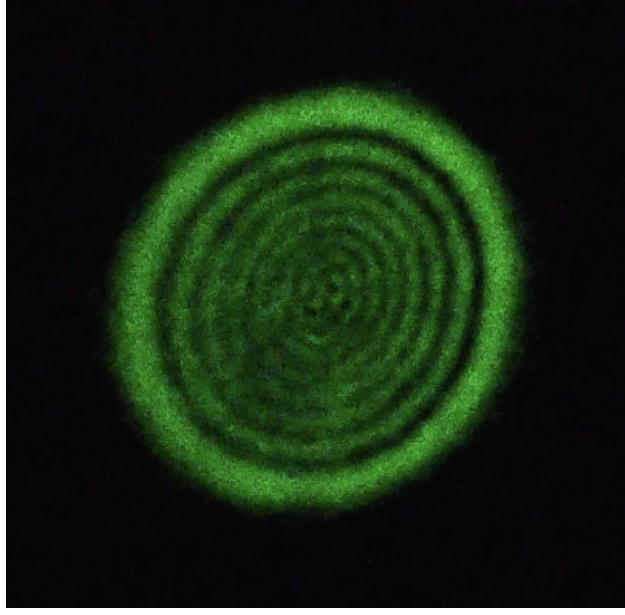


Figure 1: Example of Newton's rings in the beam spot due to the beam expander.

1. Record the beam spot with one beam blocked.
2. Record the beam spot with the other beam blocked.
3. Record the hologram produced by interference of the two beams.
4. Subtract the individual-beam intensity maps from the hologram.
5. Take the Fourier transform of the background-subtracted hologram.
6. Identify and mask one of the first-order sidebands.
7. Shift the selected sideband to the center of Fourier space.
8. Perform an inverse Fourier transform to recover the complex cross term.

With the appropriate choice of first order, the reconstructed complex field is proportional to

$$U(\mathbf{r}) = E_R^*(\mathbf{r})E_O(\mathbf{r}), \quad (13)$$

or its complex conjugate. The magnitude of U contains relative amplitude information, while its argument contains the relative phase.

2.3 Phase Shift Through Glass

For a transparent sample of thickness t and refractive index n , replacing the same geometrical path length in air gives an approximate additional optical path length

$$\Delta\text{OPL} = (n - 1)t. \quad (14)$$

The corresponding phase shift is

$$\Delta\phi = \frac{2\pi}{\lambda}(n - 1)t. \quad (15)$$

For our glass sample, $t = 0.79$ cm. A macroscopic glass plate is therefore many thousands of optical wavelengths thick. Even using a representative glass index near $n \approx 1.5$, the additional optical path corresponds to thousands of 2π phase cycles. The reconstructed phase produced by the basic DHM pipeline is wrapped into the interval $[-\pi, \pi)$, so recovering an absolute thickness from a sample this thick would require a reliable phase-unwrapping procedure and substantially better control of the wavefront and system calibration than we had available.

For that reason, our reconstruction was not expected to recover a precise height map of the glass surface. Instead, the glass was used as a deliberately obvious test object. We expected it to produce a clear discontinuity in reconstructed transmission and a corresponding phase-related change where it entered the beam. Detecting that correlated spatial change was the main experimental signature used to validate the DHM pipeline.

2.4 Laser Bandwidth and Coherence Length

The largest practical difficulty in the experiment was consistently generating a usable hologram. In addition to alignment and equipment limitations, interference visibility depends strongly on the optical path difference between the two arms. If the path difference becomes large compared with the coherence length of the source, the time-averaged cross terms wash out and the camera records mainly the two individual beam intensities.

Suppose the same source field reaches the detector along two paths with a relative time delay τ . We can write

$$E_R(t) = E(t), \quad E_O(t) = E(t - \tau). \quad (16)$$

The time-averaged intensity contains the cross term

$$\Gamma(\tau) = \langle E(t)E^*(t - \tau) \rangle, \quad (17)$$

which is the mutual coherence function for the delayed field.

Representing the field in the frequency domain gives

$$E(t) = \int A(\omega)e^{-i\omega t} d\omega. \quad (18)$$

Substituting into the correlation and time averaging eliminates unequal-frequency terms, giving

$$\Gamma(\tau) = \int |A(\omega)|^2 e^{i\omega\tau} d\omega. \quad (19)$$

Thus, the temporal coherence function is the Fourier transform of the optical power spectrum.

For a source with finite spectral width $\Delta\omega$, the envelope of $\Gamma(\tau)$ decays once the delay becomes large compared with roughly $1/\Delta\omega$. Equivalently, the interference visibility falls once the optical path difference becomes comparable to the coherence length. For a narrow optical spectrum with central wavelength λ and wavelength width $\Delta\lambda$, a commonly used order-of-magnitude estimate is

$$L_c \approx \frac{\lambda^2}{\Delta\lambda}. \quad (20)$$

The exact numerical factor depends on the definition of bandwidth and the spectral shape, so this expression should be treated as an estimate rather than an exact equality.

This coherence constraint ended up being important in two different ways. First, it explained why seemingly small changes in mirror position or beam path could dramatically change fringe visibility. Second, because the bandwidth-derived value could be compared against the actual behavior of the interferometer, it gave us a way to check whether our spectral measurement was physically consistent with the optical system.

3 System Development and Experimental Design

3.1 Design Iteration

We tested several optical geometries before arriving at the final setup. This design iteration became a central part of the project because obtaining interference alone was not sufficient: the system also needed to leave physical room for the sample, keep the optical path difference small enough for visible fringes, and produce a Fourier-domain first order that could actually be isolated from the central background and optical artifacts.

Our first major attempt was a Michelson interferometer, shown in Fig. 2.



Figure 2: Initial Michelson interferometer configuration.

The Michelson configuration produced interference, but placing an object in one arm strongly distorted the beam and made the geometry difficult to use as an imaging system. We also tested a double-mirror geometry intended to give more independent control of the object and reference paths.

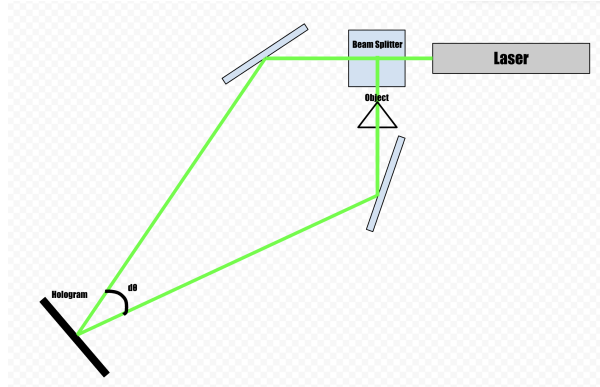


Figure 3: Alternative double-mirror geometry tested during early design iterations.

In principle, this geometry could have produced a hologram. In practice, the available mirror mounts and the limited area of the optical bench prevented us from bringing the two paths close enough together while also maintaining a sufficiently small relative beam angle. The setup therefore struggled to satisfy both constraints at once: the optical path difference had to remain small enough for interference, while the recombined beams still needed enough angular separation to move the first orders away from the DC region.

These limitations motivated a beam-splitting and recombination geometry based on a Mach-Zehnder interferometer.

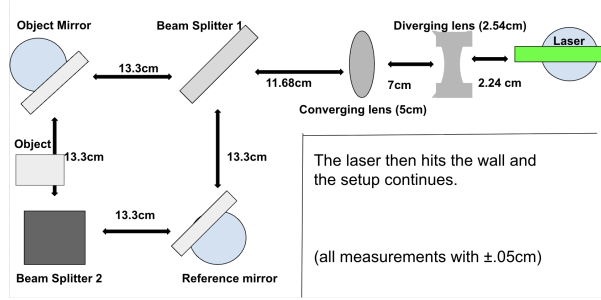


Figure 4: Mach-Zehnder geometry used as the basis for the final DHM system.

The Mach-Zehnder geometry gave us independent access to the object and reference arms and made it much easier to tune both the path lengths and the relative output angle. After repeated adjustment of mirror positions and beam overlap, we were able to obtain stable visible fringes.

Even after the interferometer itself was working, the recording geometry remained a problem. We initially placed the camera close to the outgoing optical axis, as in a conventional direct recording geometry. This produced several issues at once: low useful exposure, poor focus, limited spatial resolution, and strong specular artifacts caused by direct illumination and reflections involving the camera. In Fourier space, those artifacts spread into the same region where we expected the first-order sidebands, making the holograms difficult or impossible to reconstruct cleanly.

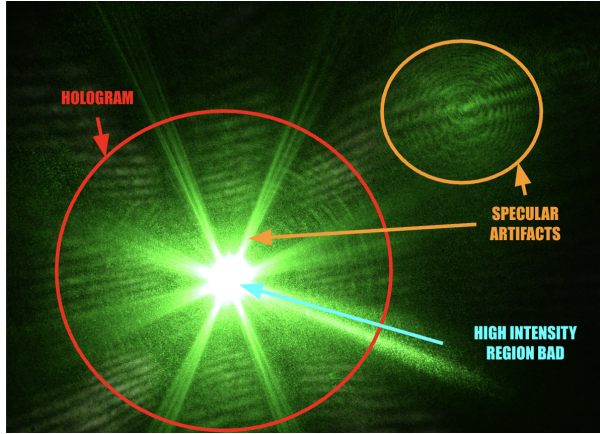


Figure 5: Imaged hologram showing specular artifacts

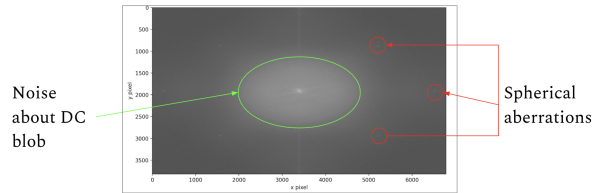


Figure 6: Fourier transform of the hologram

The final design therefore addressed two separate problems: the Mach-Zehnder geometry controlled the object-reference path difference and relative beam angle, while the camera placement was adjusted to reduce direct specular contamination. This distinction was important. The relative angle between the two recombined beams generated the off-axis spatial carrier used for Fourier separation; the camera itself was then positioned away from the strongest direct reflection path to improve the recorded hologram.

3.2 Final Mach-Zehnder Configuration

The final setup used a green laser, a diverging lens, a converging lens, two beam splitters, two mirrors, a transparent glass sample, and a camera. The two lenses acted as a simple beam expander.

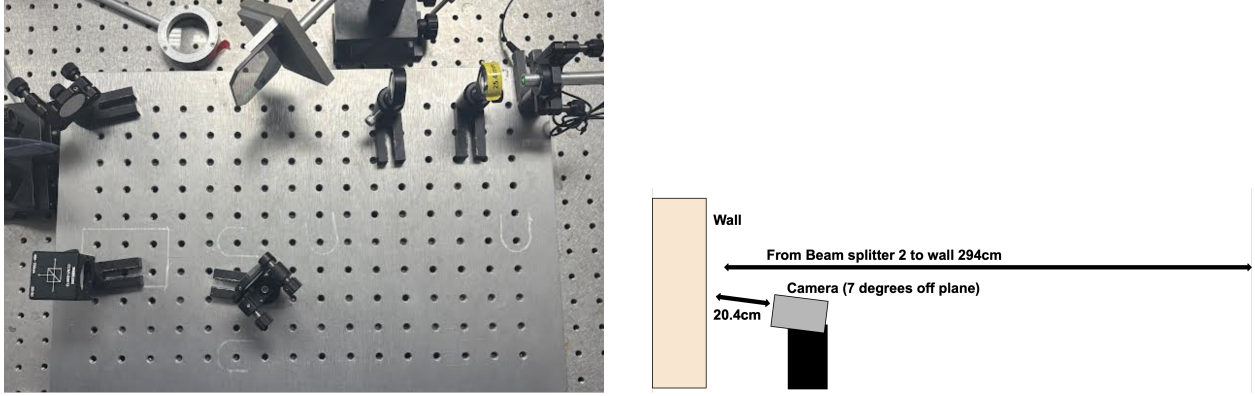


Figure 7: Final Mach-Zehnder optical layout and off-axis camera placement.

The first beam splitter created the object and reference arms, the mirrors redirected the two paths, and the second beam splitter recombined them. A small relative angle between the recombined beams generated the off-axis fringes required for Fourier-domain separation.

The camera was positioned away from the strongest direct outgoing beam path, as shown in Fig. 7. This substantially reduced the specular artifacts seen in earlier configurations and made the first-order information easier to distinguish from the central Fourier content.

Table 1: Measured component spacing in the final optical setup. Distances were measured from the previous object. The estimated ruler uncertainty was ± 0.05 cm.

Component	Distance from previous component (cm)
Diverging lens, $f = 25.4$ mm	2.24
Converging lens, $f = 50$ mm	7.00
Beam splitter 1	11.68
Object mirror	13.30
Reference mirror	13.30
Beam splitter 2	13.30
Camera distance from beam spot	20.40

The camera was positioned approximately 7° below the optical plane. The glass sample had thickness

$$t = 0.79 \text{ cm.} \quad (21)$$

3.3 Alignment Strategy

Our alignment procedure began with the beam expander. The beam spot size was checked over a long propagation distance, on the order of a few meters, to verify that the expanded beam remained approximately constant in size. If the beam spot did not noticeably expand or contract over roughly 3 m, we treated the beam as sufficiently collimated to proceed with alignment of the interferometer.

The first beam splitter was then aligned so that the heights of the incoming and outgoing beams remained consistent across both arms. We checked this by comparing the incoming beam height at the splitter with the heights of the two outgoing beams on a distant surface. The mirrors and final beam splitter were then positioned so that the two interferometer arms approximately formed

a rectangle. This geometry gave us a practical starting point for keeping the path lengths similar before fine alignment.

Fine alignment was performed by iteratively adjusting the mirrors while observing the recombined beam. The immediate goal was first to obtain spatial overlap, then to bring the optical path difference into a regime where fringes became visible. Once interference was established, one mirror was perturbed slightly to introduce a controlled angular offset between the two recombined beams. At the camera plane, this generated the off-axis fringe carrier needed to separate the first-order sidebands in Fourier space.

The alignment process was therefore not a single geometric construction but a sequence of constraints: beam collimation, equal beam height, spatial overlap, path-length matching, and finally controlled angular separation. Treating the problem in this order made the final setup substantially easier to reproduce.

3.4 Final Design Yields Improvements

The final design improved the system in several ways. First, the Mach–Zehnder geometry gave us separate control over the object and reference arms, which made path-length matching and sample insertion much more manageable. Second, the relative output angle could be tuned without completely destroying beam overlap, allowing the first-order Fourier information to move away from the central DC region. Third, shifting the camera away from the strongest direct reflection path reduced specular contamination. Finally, expanding the beam produced a larger illuminated region that was more suitable for testing a macroscopic glass sample.

The remaining limitations were not eliminated by the final design. Fringe visibility was still highly sensitive to path length and mirror adjustment, the beam expander introduced wavefront structure such as Newton’s rings, and the camera did not provide the clean dynamic range or resolution of a dedicated holographic sensor. However, the final configuration was the first one that satisfied enough of the competing constraints at the same time to support a complete reconstruction pipeline.

4 Laser Characterization and Coherence Constraints

As discussed in Subsection 2.4, one of the major bottlenecks in producing a consistent interference pattern was the requirement that the two optical paths remain sufficiently close for the source to interfere with itself. We could not find a reliable bandwidth specification for the laser in the interferometer manual, so we attempted to estimate the source bandwidth experimentally and use it to determine the scale of the coherence constraint.

This characterization served two purposes. First, it gave us a quantitative estimate of how carefully the two arms needed to be matched. Second, it provided a useful consistency check between the spectrometer measurement and the actual behavior of the interferometer.

4.1 Measured Spectrum

The measured laser spectrum is shown in Table 2. The intensity values were normalized in arbitrary units. The dominant peak occurs near 530 nm, with the measured intensity rapidly decreasing on either side.

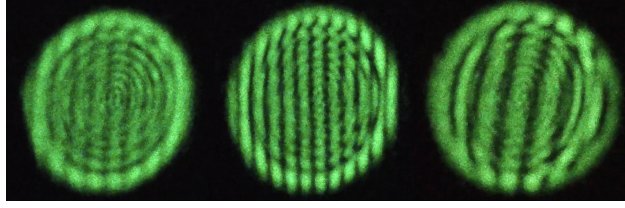


Figure 8: Representative images used during system characterization and reconstruction analysis (reference, upright glass, glass at 45 degrees from left to right)

Table 2: Measured laser spectral intensity used for the Gaussian fit. Intensities are in arbitrary normalized units.

Wavelength (nm)	Measured intensity
521	0.001582
522	0.001810
523	0.001745
524	0.001904
525	0.003275
526	0.003699
527	0.003804
528	0.007158
529	0.243808
530	0.808086
531	0.249771
532	0.019482
533	0.005042
534	0.002998
535	0.002421
536	0.001826
537	0.001745

4.2 Gaussian Bandwidth Estimate

The dominant spectral peak was fit to

$$I(\lambda) = I_{\text{bg}} + A \exp \left[-\frac{(\lambda - \mu)^2}{2\sigma^2} \right], \quad (22)$$

where I_{bg} is the background intensity, A is the peak amplitude, μ is the center wavelength, and σ is the Gaussian standard deviation.

The best-fit parameters were

$$A = 0.8048 \pm 0.0030, \quad (23)$$

$$\mu = (530.006 \pm 0.004) \text{ nm}, \quad (24)$$

$$\sigma = (0.647 \pm 0.003) \text{ nm}, \quad (25)$$

$$I_{\text{bg}} = 0.00316 \pm 0.00079. \quad (26)$$

For a Gaussian profile, the full width at half maximum is

$$\Delta\lambda_{\text{FWHM}} = 2\sqrt{2\ln 2}\sigma = 2.35482\sigma, \quad (27)$$

which gives

$$\Delta\lambda_{\text{FWHM}} = (1.524 \pm 0.006) \text{ nm}. \quad (28)$$

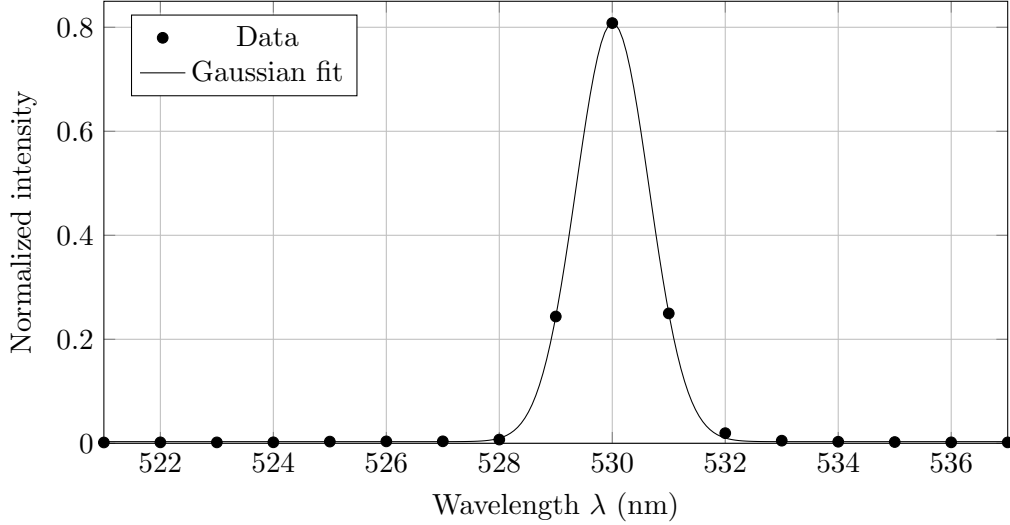


Figure 9: Measured laser spectrum with Gaussian fit. The fitted central wavelength is near 530 nm, slightly lower than the nominal 532 nm value but consistent with the observed peak of the measured spectrum.

4.3 Fit Residuals and Measurement Limitations

The residuals were calculated as

$$r_i = I_i - I(\lambda_i). \quad (29)$$

Because individual point-by-point intensity uncertainties were not recorded, we estimated the residual standard deviation from the fit,

$$s_r = \sqrt{\frac{1}{N-p} \sum_i r_i^2}, \quad (30)$$

where $N = 17$ data points and $p = 4$ fit parameters. This gave

$$s_r = 0.00292. \quad (31)$$

Using this residual scale as an effective uncertainty gives a reduced chi-squared of approximately

$$\chi_\nu^2 \approx 1.0. \quad (32)$$

This number should not be interpreted as an independent proof that the spectrum is Gaussian, because the effective uncertainty was itself estimated from the residuals. More importantly, the formal fit uncertainties on μ and σ do not include spectrometer wavelength calibration, finite wavelength sampling, or the instrument response. The largest residual occurs near 532 nm, which suggests either a non-Gaussian high-wavelength tail or instrumental/background structure.

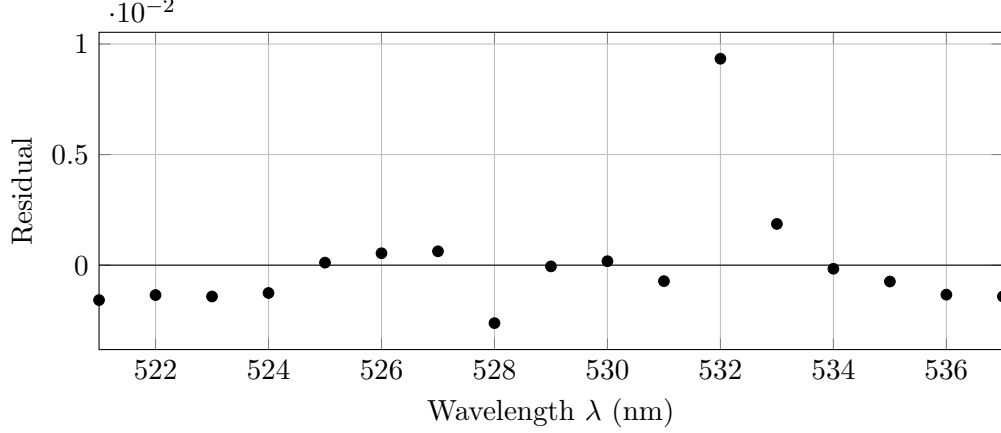


Figure 10: Residuals from the Gaussian spectral fit. Most residuals are small compared with the peak intensity. The point at 532 nm is the largest residual and likely reflects non-Gaussian spectral tail behavior or background measurement uncertainty.

4.4 Bandwidth-Based Coherence-Length Estimate

Using Eq. (20) with $\lambda = 530.006$ nm and $\Delta\lambda = 1.524$ nm,

$$L_c = \frac{(530.006 \times 10^{-9} \text{ m})^2}{1.524 \times 10^{-9} \text{ m}} = 1.842 \times 10^{-4} \text{ m} = 0.184 \text{ mm}. \quad (33)$$

If only the formal fit uncertainties are propagated,

$$\left(\frac{\sigma_{L_c}}{L_c}\right)^2 = \left(2\frac{\sigma_\lambda}{\lambda}\right)^2 + \left(\frac{\sigma_{\Delta\lambda}}{\Delta\lambda}\right)^2, \quad (34)$$

giving

$$L_c = (0.184 \pm 0.001) \text{ mm}. \quad (35)$$

Taken literally, this is a very short coherence length compared with the centimeter-scale dimensions of the interferometer. It is nevertheless consistent with one important qualitative observation: small changes in mirror angle, component placement, or path length often caused fringe visibility to change dramatically.

4.5 Consistency Check Against the Interferometer

The nominal laser wavelength was 532 nm, while the fitted peak wavelength was

$$\mu = (530.006 \pm 0.004) \text{ nm}. \quad (36)$$

The difference is

$$\Delta = 532.000 - 530.006 = 1.994 \text{ nm}. \quad (37)$$

If the formal fit uncertainty were the only uncertainty in the measurement, this would appear to be a very large discrepancy. In practice, however, the nominal 532 nm value is not a precision calibration standard, and the measured spectrum is also affected by spectrometer calibration and finite wavelength sampling. We therefore treat the fitted center as a description of the measured spectrum rather than as a precision measurement of the laser's true absolute wavelength.

A useful consistency check comes from comparing this estimate with the scale introduced by the glass. The plate had a thickness of roughly 0.79 cm. If we take a representative refractive index $n \approx 1.52$, the portion of the beam transmitted through the glass would acquire an additional optical path of approximately

$$\Delta\text{OPL} \approx (1.52 - 1)(0.79 \text{ cm}) \approx 4 \text{ mm}. \quad (38)$$

This is much larger than 0.184 mm. We nevertheless observed visible interference with the partially inserted glass sample. That observation raised an immediate consistency question, although it does not by itself establish a millimeter-scale lower bound on coherence length because the glass covered only part of the beam and we did not perform a calibrated fringe-visibility-versus-delay measurement.

For that reason, we do not treat 0.184 mm as a reliable physical measurement of the laser coherence length. It is better interpreted as the coherence scale implied by our measured spectral width under a simplified model. The discrepancy between that scale and the qualitative interferometer behavior points to limitations of the coarse spectral sampling, spectrometer response, bandwidth definition, and the lack of a direct coherence-length scan.

This mismatch became a useful part of the project rather than simply an incorrect result. It showed why the spectral fit and the interferometer observations needed to be compared critically, and why the very small formal fit uncertainty should not be confused with the much larger uncertainty in the physical coherence-length estimate.

5 Hologram Reconstruction and Qualitative Validation

Because the final setup was still limited by alignment stability, camera quality, and wavefront artifacts, we did not expect it to resolve small microscopic features with metrology-level accuracy. The immediate question was more basic: could the system recover a spatial feature that was actually tied to a physical object?

To answer that, we deliberately chose a simple “smoking gun” sample. A diffuse piece of glass was positioned so that it covered only part of the beam spot. This created a large, easy-to-identify boundary in the optical field rather than a subtle microscopic feature. The uncovered portion of the beam should remain comparatively bright, while the portion passing through the diffuse glass should experience both additional optical phase and reduced transmitted intensity. If the DHM pipeline was functioning correctly, a reconstructed transition should appear near the physical boundary of the glass.

5.1 Fourier-Order Isolation

We then ran the full DHM reconstruction pipeline on the recorded holograms. To reduce temporal noise and make the first-order peaks easier to identify, the Fourier-domain data were averaged across the available frames. This was useful because the raw holograms contained visible speckle, nonuniform illumination, and fringe-contrast variations that made the sidebands difficult to isolate from a single frame.

The first orders were then selected using circular masks in Fourier space. The mask location and radius were chosen to capture as much of the sideband as possible without including excessive DC leakage or nearby artifacts. Because the reconstructed complex field depends strongly on which Fourier order is selected and how much of it is retained, this mask choice remained one of the largest reconstruction-related systematic uncertainties.

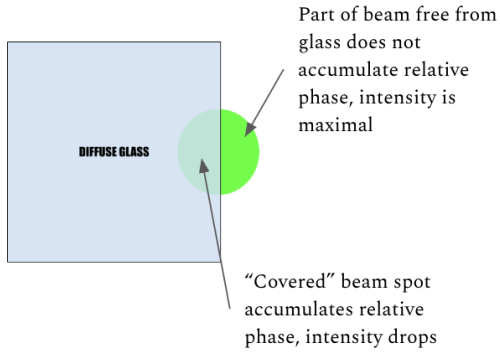


Figure 11: Diffuse glass partially covering the beam spot.

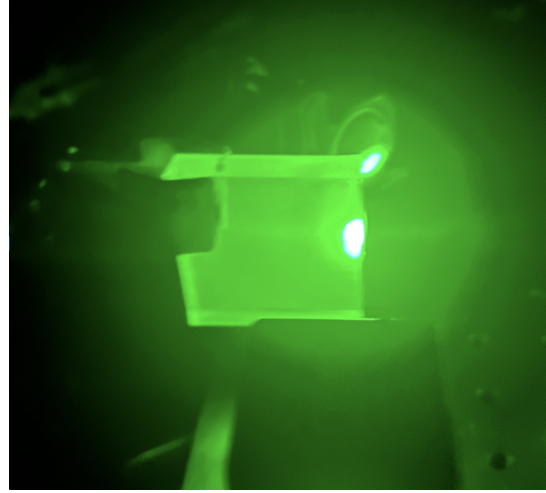


Figure 12: Experimental realization of the diffuse glass setup.

Figure 13: “Smoking gun” diffuse glass test used to validate that the DHM pipeline was reconstructing physical structure.

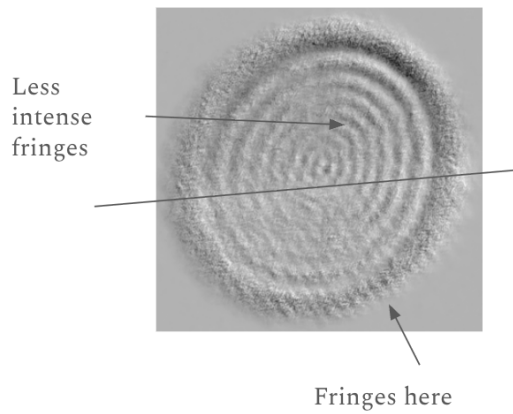


Figure 14: One frame of the recorded hologram

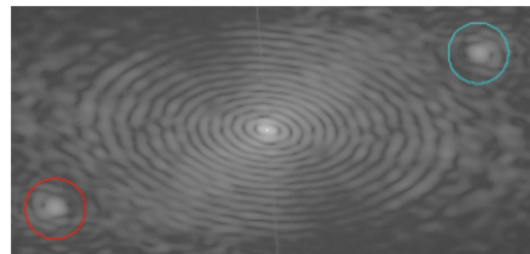


Figure 15: Masked first-order diffraction peaks.

Figure 16: Isolation of the first diffraction orders used for reconstruction.

After masking, the selected sideband was shifted toward the Fourier origin and inverse transformed to obtain a complex reconstructed field. The field magnitude was used to form intensity-related maps, while the complex argument was used for the phase reconstruction.

5.2 Relative Transmission

After reconstruction, the intensity maps for the object and reference conditions were computed. The relative transmission was estimated using

$$\frac{I_{\text{obj}}}{I_{\text{ref}}}.$$

This ratio helps normalize out part of the nonuniform beam profile and makes the effect of the diffuse glass easier to distinguish from the underlying illumination pattern. A threshold was applied to suppress low-intensity regions outside the useful beam area, since the ratio becomes unstable wherever the reference intensity approaches zero.

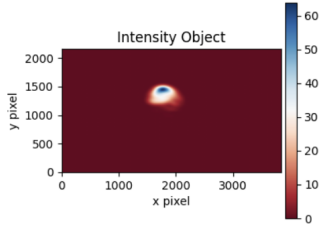


Figure 17: Object intensity.

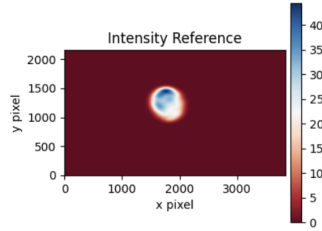


Figure 18: Reference intensity.

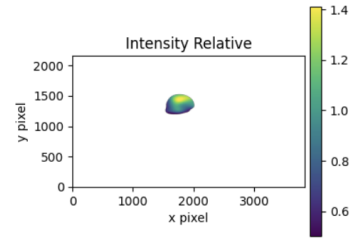


Figure 19: Relative transmission map.

Figure 20: Reconstructed intensity fields and relative transmission estimate.

The reconstructed transmission map shows a clear suppression in the portion of the beam passing through the diffuse glass. More importantly, the location and orientation of that suppression approximately match the physical boundary of the glass in the beam. This is the first major validation step: the dominant feature in the reconstruction tracks a known object-space feature rather than appearing at an arbitrary location in the image.

The result should still be interpreted qualitatively. The exact transmission values depend on imperfect beam overlap, fringe contrast, threshold choice, and the Fourier mask used during reconstruction. The useful result is therefore the spatial correspondence, not a precision transmission coefficient.

5.3 Phase-Related Reconstruction

The reconstructed phase field was also converted into a relative optical-path or height-like map. The physical idea is that light traveling through a material with a different refractive index accumulates additional optical phase. Under appropriate assumptions and with reliable phase unwrapping, that phase can be converted into a relative thickness or surface-height variation.

Our glass sample is far too thick, and our phase reconstruction too noisy, for that conversion to be interpreted as an accurate physical height measurement. For this reason, the phase-derived map is used here primarily as a visualization of spatial phase structure rather than as a calibrated surface profile.

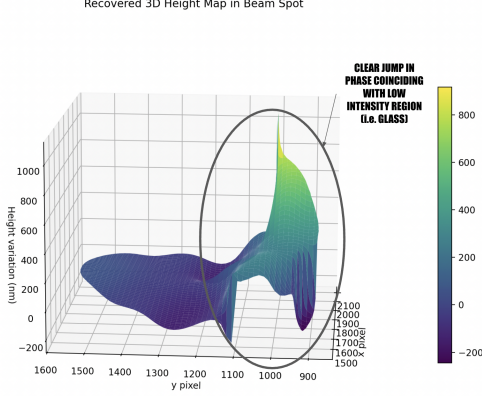


Figure 21: Phase-derived relative height-like map.

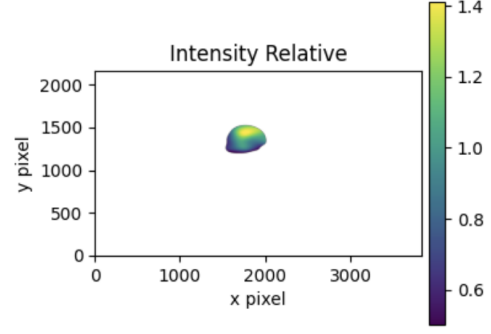


Figure 22: Relative transmission map used for comparison.

Figure 23: Comparison between the phase-derived spatial structure and relative transmission.

Despite substantial low-frequency distortion and noise, the phase-related transition occurs in approximately the same region as the transmission change. The two maps are not expected to match pixel-for-pixel, since scattering, phase wrapping, imperfect masking, and wavefront curvature affect them differently. However, their shared large-scale boundary provides a second, independent indication that the reconstructed feature is associated with the glass.

Overall, the diffuse-glass test converts the reconstruction from a purely visual demonstration into a basic validation experiment. The setup remains too limited for quantitative phase microscopy, but the reconstructed transmission and phase-related fields both respond to the known position of the sample. That is the main evidence that the Fourier-filtered DHM pipeline is recovering physical spatial information rather than only numerical or optical artifacts.

6 Profile-Based Quantitative Validation

The two-dimensional reconstructions were useful for visual validation, but we also wanted a more compact way to compare the phase-related and transmission features. The goal was not to extract a high-precision glass thickness. Instead, we used the reconstructed maps to test whether the two observables contained a transition at approximately the same sample boundary.

To reduce local speckle, masking artifacts, and pixel-scale noise, one axis of each reconstructed map was collapsed by taking the mean along that direction. This produced one-dimensional profiles of accumulated relative phase and relative intensity as functions of normalized vertical position across the beam.

If the glass boundary were reconstructed perfectly, each profile would resemble a sharp step: one approximately constant value in the uncovered region and another in the glass-covered region. In the actual data, the transition is broadened by the finite beam size, imperfect edge localization, Fourier-mask width, wavefront distortions, and reconstruction noise. We therefore modeled the transition with a sigmoid,

$$f(x) = y_0 + a \frac{1}{1 + e^{-c(x-x_0)}},$$

where y_0 is the baseline, a is the change across the transition, c controls the steepness, and x_0 gives the approximate transition location.

For the relative-intensity fit, the sigmoid was constrained so that the fitted ratio remained physically reasonable. This was necessary because the ratio $I_{\text{obj}}/I_{\text{ref}}$ becomes unstable in regions where the reconstructed reference intensity is very small.

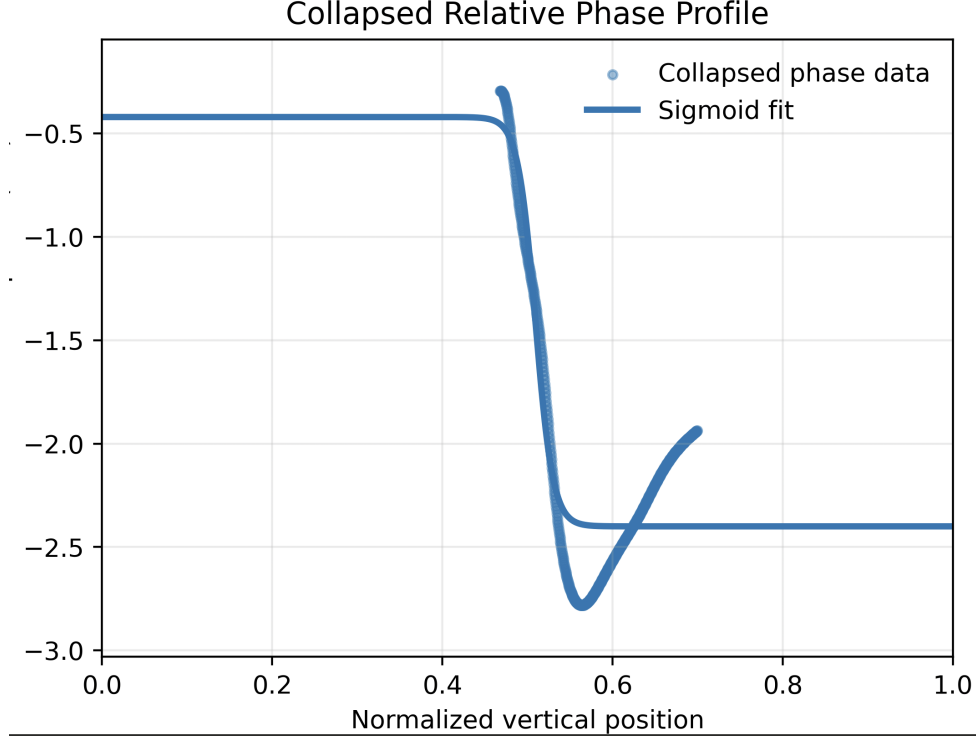


Figure 24: Collapsed accumulated relative phase profile fit to a sigmoid.

The phase fit is useful because the glass should introduce additional optical path length and therefore an accumulated phase change. After correcting the sign convention of the reconstructed phase, the phase profile shows a transition complementary to the relative-intensity profile. The relative intensity decreases in the region where the beam passes through the diffuse glass, while the phase changes over approximately the same region.

This correspondence is more useful than either fit in isolation. A numerical artifact could produce structure in one reconstruction, but the appearance of related transitions in two separately interpreted quantities makes the sample-based explanation more plausible.

The raw chi-squared-like residual measures from the sigmoid fits were

$$\chi_{\text{phase}}^2 = 30.845683896179008,$$

and

$$\chi_{\text{intensity}}^2 = 15.375314811674373.$$

These values should not be interpreted as rigorous statistical chi-squared goodness-of-fit measurements because we do not have independently measured point-by-point uncertainties for the collapsed phase or intensity data. In particular, without a defined variance for each point, a reduced chi-squared or statistical p -value would not be meaningful. Here, the values are best treated as residual diagnostics associated with the chosen sigmoid model.

The more important quantitative outputs are the existence, direction, and approximate location of the transitions. Both collapsed profiles show a smooth change near the region where the diffuse

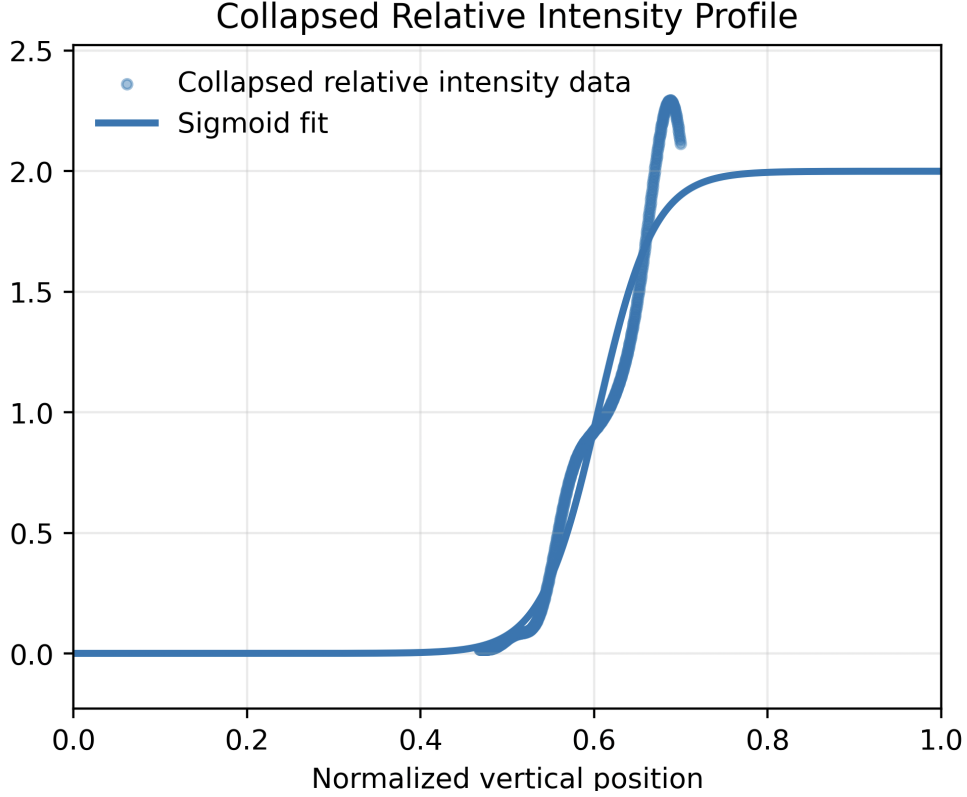


Figure 25: Collapsed relative intensity profile fit to a sigmoid.

glass enters the beam, and the two profiles change in complementary directions. This supports the same conclusion as the two-dimensional images: the reconstruction is responding to a real sample feature, even though the setup is not precise enough to recover an accurate absolute thickness or surface profile.

7 Experimental Limitations and Uncertainty

The dominant uncertainties in this project were not all well described by a single numerical error bar. Some came from ordinary measurement noise, while others came from optical alignment, wavefront quality, camera geometry, and the reconstruction choices made in Fourier space. For that reason, it is useful to separate the limitations into random and systematic contributions.

7.1 Random and Repeatability Limits

The main random or repeatability-related uncertainties were:

- finite camera pixel resolution,
- small frame-to-frame changes in the interference pattern,
- uncertainty in locating and sizing the Fourier first-order mask,
- intensity noise in the spectral measurement, and
- ruler uncertainty of approximately ± 0.05 cm for optical component spacing.

These effects mainly changed the detailed appearance of a reconstruction from one frame or masking choice to another. Averaging across frames and collapsing the reconstructed maps into one-dimensional profiles reduced some of this variability, but did not remove it completely.

7.2 Systematic Limits

The dominant systematic effects were more important for the overall image quality:

- mismatch between the two optical path lengths,
- imperfect beam overlap at the second beam splitter,
- specular artifacts when the camera was too close to the strongest outgoing beam direction,
- residual wavefront curvature and Newton’s rings from imperfect beam expansion,
- nonuniform scattering from the diffuse glass sample,
- camera saturation and exposure dependence, and
- sensitivity of the reconstructed field to the exact Fourier-order mask.

The bandwidth-derived coherence-length estimate of 0.184 mm initially suggested that the two paths needed to be matched with sub-millimeter precision. The physical ruler uncertainty on individual component positions was already about 0.5 mm, so absolute bench measurements alone were not precise enough to guarantee interference. In practice, the system therefore had to be aligned iteratively by directly observing fringe visibility rather than by relying only on measured component spacing.

At the same time, as discussed in the laser-characterization section, the qualitative interferometer behavior was not cleanly explained by the 0.184 mm estimate. Because the glass only partially covered the beam and no calibrated visibility-versus-delay scan was performed, the experiment cannot determine the true coherence length from that observation alone. The formal uncertainty from the Gaussian spectral fit therefore understates the actual uncertainty in the physical coherence-length estimate. The main lesson is not that the coherence length was known to high precision, but that coherence was a dominant system-level constraint and had to be managed experimentally.

7.3 Implications for the Reconstruction

These limitations determine how strongly the final images can be interpreted. The reconstructed intensity and phase maps are sufficient to test whether the glass produces a repeatable spatial feature, but they are not sufficient for a high-accuracy measurement of absolute sample thickness or surface profile. In particular, quantitative phase metrology would require better wavefront calibration, a better-characterized light source and sensor, more stable optomechanics, and a more rigorous phase-unwrapping procedure.

For this reason, the strongest claim supported by the experiment is a validation claim rather than a precision-metrology claim: the interferometer and reconstruction pipeline recovered spatial structure that followed the physical location of the sample in both the transmission and phase-related reconstructions.

8 Results and Interpretation

The final Mach–Zehnder system produced visible, stable interference fringes and supported a complete Fourier-domain reconstruction workflow. The reference, upright-glass, and angled-glass conditions produced visibly different holograms, and the reconstructed field changed in the region where the glass intersected the object beam.

The strongest result is not simply that an interference pattern was observed. The project progressed from raw fringe generation to a sample-dependent reconstruction: a first-order sideband was isolated, inverse transformed into a complex field, and used to generate transmission and phase-related maps. In both maps, the dominant large-scale transition appeared near the physical boundary of the diffuse glass.

This supports the central claim of the project: the final setup was not merely recording an ordinary shadow or a camera artifact. Information about the sample was encoded in the object-reference interference pattern and recovered after Fourier filtering. The complementary behavior of the collapsed intensity and phase profiles provides an additional cross-check that the reconstructed structure is associated with the sample.

At the same time, the system did not reach the regime of quantitative phase microscopy. The reconstructed phase was affected by wrapping, low-frequency wavefront distortion, Fourier-mask sensitivity, and the large thickness of the glass. The bandwidth-derived coherence-length estimate also raised a consistency question when compared with the qualitative interferometer behavior, reinforcing that the spectral measurement did not provide a precise physical coherence-length determination.

Taken together, the result is best described as a successful proof-of-principle DHM system with experimentally identified limitations. The project demonstrated the full chain from optical design and alignment through hologram acquisition, Fourier reconstruction, and sample-based validation, while also showing which parts of the system would need to be improved before attempting microscopic or metrology-grade measurements.

9 Conclusion

We designed, built, and iteratively refined an off-axis DHM system based on a Mach–Zehnder interferometer. Earlier configurations either failed to produce stable interference, could not accommodate the sample geometry, or produced Fourier-domain first orders that were too strongly contaminated by DC leakage, wavefront structure, and specular artifacts. The final system combined a Mach–Zehnder geometry with controlled beam-angle offset, beam expansion, and modified camera placement to produce usable holograms.

A Gaussian fit to the measured laser spectrum gave

$$\mu = (530.006 \pm 0.004) \text{ nm}, \quad \Delta\lambda_{\text{FWHM}} = (1.524 \pm 0.006) \text{ nm}. \quad (39)$$

Using the standard bandwidth-based estimate gave

$$L_c = (0.184 \pm 0.001) \text{ mm}. \quad (40)$$

However, the qualitative behavior of the interferometer was not cleanly explained by such a short value. Because the glass covered only part of the beam and we did not perform a calibrated fringe-visibility-versus-delay scan, that observation cannot determine the true coherence length by itself. We therefore interpret the 0.184 mm value as a model- and instrument-limited estimate derived from the measured spectrum rather than a high-precision physical measurement. The discrepancy

was still informative because it forced a consistency check between the spectral characterization and the behavior of the optical system.

The final validation used a diffuse piece of glass partially covering the object beam. Because the sample was macroscopic and introduced many phase wraps, the appropriate target was not an absolute thickness measurement. Instead, we looked for a reconstructed spatial transition that followed the physical glass boundary. The relative-transmission map showed clear suppression in the glass-covered region, while the phase-related reconstruction changed over approximately the same region. Collapsed one-dimensional profiles showed complementary transitions, providing additional evidence that the recovered feature was sample related.

Overall, the project successfully demonstrated the main elements of digital holographic imaging with the available Physics 5CL equipment: interferometric encoding, off-axis Fourier separation, complex-field reconstruction, and qualitative sample validation. Just as importantly, the project identified the main limitations preventing quantitative microscopy, including source characterization, path-length sensitivity, wavefront distortion, camera artifacts, mechanical stability, Fourier-mask dependence, and phase unwrapping.

Future improvements would include a better-characterized narrow-linewidth laser, a higher-quality CCD or CMOS sensor, improved optomechanical mounts, a calibrated microscope objective, cleaner beam expansion, automated Fourier-order selection, and a robust phase-unwrapping and background-phase correction pipeline. With those improvements, the same general system could be extended from macroscopic proof-of-principle imaging toward true microscopic quantitative phase imaging.

10 References

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11 Appendix: Notes on Reproducibility

A future group attempting to reproduce this system should treat alignment as a sequence of constraints rather than assembling the full interferometer at once. The beam expander should first be adjusted until the expanded spot remains approximately constant over a long propagation distance. The first beam splitter should then be aligned so that both arms remain at the same height, followed by placement of the mirrors and recombining beam splitter in an approximately rectangular geometry.

The next priority is spatial overlap at the recombiner. Once the two beams overlap, the path lengths should be adjusted while directly monitoring fringe visibility. Ruler measurements are useful for bringing the two paths into the same approximate range, but the final adjustment should be based on the interference itself because the relevant optical path difference is more sensitive than the mechanical spacing measurements.

After stable fringes are obtained, a small relative angular offset should be introduced between the object and reference beams so that the first-order sidebands separate from the central Fourier content. The camera should be positioned to avoid the strongest direct reflection and saturation artifacts while still recording the overlapping region of the two beams.

For reconstruction, object-only, reference-only, and two-beam hologram images should be recorded without changing the exposure or optical alignment. Multiple frames should be collected when possible so that temporal averaging can be used. The Fourier mask should be documented because its position and radius can noticeably change the reconstructed field.

Finally, a simple partially inserted sample such as the diffuse glass used here is useful before attempting a microscopic target. A large known boundary provides a direct validation test: the reconstructed transmission and phase-related maps should change in approximately the same physical region. Only after that basic sample correspondence is established would it make sense to move toward thinner samples, phase unwrapping, and quantitative thickness measurements.